

Frederick Khasanto

PhD Student in Intelligent Transportation | HKUST (Guangzhou) | Advised by Prof. Benedict Jun MA

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) 2026 - present

PhD in Intelligent Transportation

Advisor: Prof. Benedict Jun MA

The Chinese University of Hong Kong, Shenzhen Sept 2022 - Jun 2026

B.Sc. in Data Science and Big Data Technology | CGPA 3.79 / 4.00 | MGPA 3.78 / 4.00

Relevant coursework: Mathematical Statistics, Optimization, Advanced Machine Learning, Reinforcement Learning, Stochastic Processes & Simulation, Natural Language Processing, Deep Learning and Applications

RESEARCH EXPERIENCE

Research Assistant - Offensive Speech Detection Project, CUHK(SZ) Sep 2025 - present

- Conducted literature review on transformer-based hate-speech detection under Prof. Benyou Wang research group.
- Collected existing datasets for supervised classification.

Summer Research Intern - VLAA Lab, University of California, Santa Cruz Jul 2025 - Aug 2025

- Reviewed vision-language model literature, including LLaVA and LLaVA-CoT, and multimodal architectures.
- Summarized findings under the supervision of Hardy Chen and Prof. Cihang Xie.

SELECTED PROJECTS

Diffusion Models for X-ray Image Generation - Deep Learning Project Oct 2025 - Dec 2025

- Implemented a DDPM for medical X-ray image generation in PyTorch.

KnowYourBody - AI Practicum Project Oct 2025 - Dec 2025

- Developed a web app integrating LLM APIs for educational and personalized health analytics.

SmartBot: Cloud-based Clinic Appointment Chatbot Oct 2024 - Dec 2024

- Developed an appointment scheduling chatbot using AWS Cloud Functions.

TEACHING

Undergraduate Student Teaching Fellow - STA2002 Probability and Statistics II Sep 2025 - present

- Led tutorials on point estimation, interval estimation, hypothesis testing, and order statistics; conducted a midterm review session.

Undergraduate Student Teaching Fellow - STA2002 Probability and Statistics II Jan 2025 - May 2025

- Led weekly tutorials on Chi-square tests, ANOVA, linear regression, and likelihood ratio tests; designed two homework assignments.

Undergraduate Student Teaching Fellow - DDA3010 Applied Machine Learning Jan 2024 - May 2024

- Held office hours to assist students in understanding course materials.

Undergraduate Student Teaching Fellow - CSC1001 Introduction to Programming Methodology Sep 2023 - Dec 2023

- Led weekly Python tutorials, created problem sets and live-coding examples, and graded assignments with feedback.

LEADERSHIP AND SERVICE

Peer Advisor Group Leader - School of Data Science, CUHK(SZ) Sep 2024 - Jul 2025

- Coordinated meetings between academic advisors, peer mentors, and freshmen; mentored first-year students on academic planning and campus adaptation.

Vice Chair, Academic and Career Support Dept. - International Students' Association, CUHK(SZ) Aug 2023 - Jul 2024

- Organized mentoring programs, academic talks, alumni panels, company sharing sessions, and site visits.

University Library Student Assistant - CUHK(SZ) Feb 2023 - May 2024

- Assisted with material cataloging and logistical support for workshops and student events.

AWARDS AND SCHOLARSHIPS

- Dean's List, School of Data Science (2023, 2024, 2025)
- Excellent Student Leader Award - University Honors (2025)
- Outstanding Leadership Award - Muse College (2024, 2025)
- Shenzhen Universiade International Student Scholarship (2024)
- 3rd Prize - CUHK(SZ) Programming Contest (2023)
- Guangdong Government Outstanding International Student Scholarship (2022, 2023)
- Full Tuition and Accommodation Scholarship (2022 - present)

TECHNICAL SKILLS

Programming: Python, R, C++, SQL

Frameworks: PyTorch, Scikit-learn, Pandas, NumPy

Tools: Git, LaTeX

Languages: English (Fluent), Chinese (Limited Working Proficiency)

RESEARCH INTERESTS

Intelligent Transportation, Machine Learning, Data Science, Natural Language Processing, Multimodal AI